MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCCFS202 - COMPUTER SECURITY

Duration: 1.5 hrs MaxiumMarks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attain ability.

Part A

Short Answer Type Questions
Answer **any 10** Questions
Each question carries **1** mark

1. Explain Computer Security. [Understand][1]
2. Discuss about malware. [Understand][1]
3. List the three components of the CIA triad. [Remember][1]
4. Compare symmetric and asymmetric encryption. [Analyse][2]
5. Define IP spoofing. [Understand][1]
6. Expand DMZ in the context of firewall architecture. [Understand][1]
7. Identify the IDS type suitable for detecting unauthorized changes to system files on a server. [Apply][3]
8. Compare site-to-site VPN and remote access VPN. [Analyse][2]
9. Examine the purpose of TLS handshake protocol. [Analyse][2]
10. Explain about the purpose of database auditing. [understand][1]
11. Contrast static and dynamic analysis in mobile app security testing.[Analyse][2]
12. Explain the term multitenancy in cloud security. [understand][1]

[10x1=10]

Part B

Short Essay Type Questions
Answer **any 4** Questions
Each question carries **5** marks

13. Compare Mandatory Access Control (MAC) and Discretionary Access Control (DAC). [Analyse][2]
14. Discuss about the man-in-the-middle Attack and how can it be prevented. [Understand][1]
15. Differentiate between signature based detection and anomaly based detection. [Analyse][2]
16. Compare the functionalities of Authentication Header (AH) and Encapsulating Security Payload (ESP). [Analyse][2].
17. Explain about Cross-site Request Forgery (CSRF) and explain how it works with an example. [Understand][1]
18. Examine the importance of database backup and recovery with an example.

[4x5=20]

Part C

Essay Type Questions

Answer **any 2** Questions

Each question carries **10** marks

19. Discuss about the Bell-LaPadula Model. [Understand][1]
20. Categorize the different SQL injection attacks with examples and prevention techniques. [Analyse][2]
21. Illustrate the layered approach to cloud security.[Analyse][2]

[2x10=20]